MANTHAN DHANRAJ

Full-Stack & AI Developer | Computer Science Engineering Student

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Professional Summary

Computer Science Engineering student and hands-on builder who ships real, deployed products. Skilled in full-stack development with Python/Flask and Node.js/Express, machine learning on real-world datasets, and computer-vision pipelines. Comfortable owning projects end-to-end — from architecture and model training to production deployment.

Skills

Languages: Python, C, JavaScript, SQL
Web & Backend: Flask, Node.js, Express.js, REST APIs, HTML5, CSS3, SQLite, PostgreSQL (Supabase)
AI / ML: scikit-learn, Random Forest, MediaPipe (pose estimation), faster-whisper (STT), Ollama (local LLMs)
Data & Analysis: pandas, NumPy, feature engineering, model evaluation
Tools & Deployment: Git, GitHub, VS Code, Render, CUDA/GPU, Tkinter, MS Office
Core CS: Data Structures & Algorithms, OOP, File I/O

Projects

AthletixAI — AI Sports Biomechanics Platform | *Flask, MediaPipe, SQLite* **Live**

- Built and deployed a live web app that analyzes athletic technique across 48 sports using AI-powered biomechanics scoring.
- Engineered a computer-vision pipeline with MediaPipe 33-keypoint pose analysis to evaluate movement and generate per-sport performance feedback.
- Developed full-stack features: multi-user persistence, secure email (Gmail/SSL), an admin panel, and a premium charcoal/gold design system.
- Deployed to production on Render; owned the complete stack from backend architecture to UI. Live at athletixai.onrender.com.

JARVIS OS — AI Voice Assistant for Windows 11 | *Python, faster-whisper, Ollama, CUDA* **v0.7.0**

- Building a JARVIS-style AI operating-system assistant (77 Python modules) with full voice control in Hindi, English, and Hinglish.
- Implemented offline speech-to-text (faster-whisper with CUDA acceleration), wake-word detection, and risk-based command confirmation gating.
- Integrated a local LLM (Ollama) for Hinglish intent understanding with SQLite-backed memory and context awareness.
- Built Windows automation, browser/coding/gaming modes, safe file management, and a Tkinter GUI dashboard, all with safety-first, env-gated power controls.

SOHelioraX — Solar Flare Nowcasting (ISRO BAH 2026) | *Flask, scikit-learn* **Team Lead**

- Led a 3-person team to build an ML-powered web app that forecasts solar flares using real Aditya-L1 satellite data (SoLEXS + HEL1OS).
- Trained a Random Forest model achieving ~97.7% cross-validated accuracy across 739 days of space-weather data.
- Built a live dashboard with real-time alerts and email subscriptions; authored a 23-page technical report and pitch deck.

Unifind — University Discovery Platform | *Node.js, Express, PostgreSQL (Supabase)* **In Progress**

- Building a web app to help students find and explore Indian universities, their courses, and fee structures.
- Designed a normalized Supabase PostgreSQL schema and a Node.js/Express REST API using LEFT JOIN aggregation to return courses as JSON.

DSA Implementation Library | *C* —

Implemented core data structures (stack, queue, linked list) and search/sort algorithms with documented time/space complexity.

Education

Sharda University, Greater Noida, India **2025 – 2029**

Bachelor of Technology in Computer Science Engineering — CGPA: 8.28 / 10

Achievements & Certifications

OpenAI Codex Hackathon (NamasteDev): Finished in the top 30% of 2,989 participants for the submitted project.
Certifications: Python Programming • Android App Development • AI Tools & ChatGPT Workshop • Google Analytics Fundamentals • Digital Marketing Fundamentals • Project Management Fundamentals

Activities & Interests

Build and ship side projects, solve DSA problems on online judges, and explore open-source repositories to learn production code structure.